

Spring—25

2. a) What is a system? Discuss different types of information systems and their roles in organizations. 5 CO1
- b) What is the Systems Development Life Cycle (SDLC), and how does it help in building an information system? Explain the key phases of the SDLC and provide an example of how each phase might be applied in the development of an e-commerce platform. 5 CO1
- OR (of 2b)
- ✓ Which SDLC model would you adopt for developing an e-commerce platform that requires small and there are no uncertain requirements? Justify your answer reasoning with the advantages and disadvantages of your model. 5

Spring—24

2. a) Discuss Rapid Application Development (RAD) and Joint Application Development (JAD) methodologies with their advantages and disadvantages. 5 CO1
- OR (of 2a)
- What is SDLC? Briefly describe the different phases of SDLC.
- b) A software development company currently offered to develop following 5 CO2 projects:
- i. A new accounting software for a client. The client has provided detailed requirements upfront and expects the software to be delivered in a short period of time.
 - ii. Another project is a new social networking platform. The company wants to adapt to changing market needs quickly and release new features iteratively to gather user feedback. Agile
- Above the scenario which SDLC model would adopt for developing both projects? Justify your answer.

Autumn—23

- 2.a) Which SDLC model would you adopt for developing an e-commerce platform that requires small and there are no uncertain requirements? Justify your answer reasoning with the advantages and disadvantages of your model. 5 CO1 U
- Or,
- Suppose you want to develop an online learning platform. It develops modules on various subjects, enabling instructors to create and deliver content, and students to engage in interactive learning. Regular feedback from educators helps refine the platform's features and usability. Which SDLC model will be more suitable for the above learning platform? Justify your answer
- b) Distinguish between Methodologies, Models, Tools and Techniques with example. 5 CO1 U

Spring = 25

Answer to the Ques. No - 2(a)

System: A system is a set of components working together to achieve a goal.

Example: Education system = admission + classes + exams + results.

Different types of information systems:

① Transaction processing system (TPS)

⇒ It handles day to day routine operations.

⇒ It is used at the operational level where tasks are highly structured.

⇒ Example: ATM transactions, supermarket billing^{etc}.

⇒ It supports daily operations.

② Management information system (MIS)

⇒ It is a set of information processing functions.

⇒ It provides summarized information for middle managers.

⇒ It helps in planning and controlling.

⇒ Example: Budgeting system, Sales Management system etc.

③ Decision Support system (DSS)

- ⇒ It helps managers in decision-making by analyzing data.
- ⇒ Use of analytical and simulation models.
- ⇒ Some artificial intelligence tools may be used.
- ⇒ Assist in problem solving and decision-making

Example: Financial planning system.

④ Executive Support system (ESS)

- ⇒ Give top-level executives strategic information and dashboards.
- ⇒ ESS rely on the information generated by TPS, MIS and DSS.
- ⇒ It present graph and data from many sources

Example: External database.

⑤ Enterprise system

- It is known as enterprise resource planning (ERP)
- It helps to improve coordination and decision making.

⑥ Supply chain management system (SCM)

- ⇒ It manages flow of goods, services and information from suppliers to customers.
- ⇒ It is widely used by manufacturing and distribution firms to connect to suppliers.
- ⇒ The ultimate goal is to get the right amount of the products.

Example: Retail customer, Manufacturer etc.

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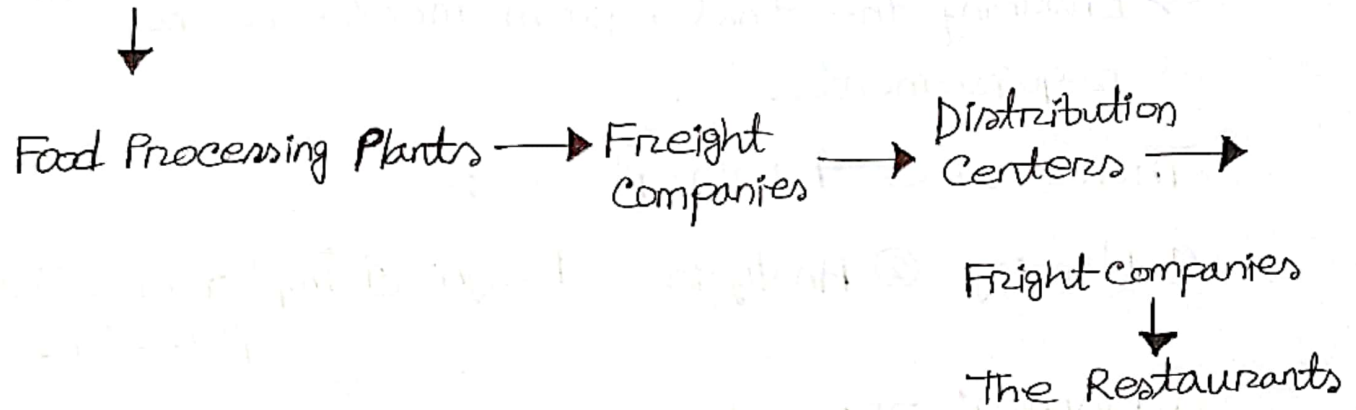


Fig: Supply chain.

⑦ Customer Relationship Management System (CRM)

- ⇒ Manages customer data, sales, feedback, and marketing campaigns.

Example: Salesforce CRM, Hubspot CRM etc.

Ans. to the Ques. No - 2(b)

Systems Development Life cycle (SDLC):

The SDLC is a step by step process used to plan, design, build, test and maintain an information system. It helps developers and managers by:

- ⇒ Giving a structured approach so nothing is missed.
- ⇒ Reducing risk and controlling time and cost.
- ⇒ Ensuring the final system meets user requirements.

There are 4 main phases:-

① Planning ② Analysis ③ Design ④ Implementation & operation.

① Planning: The planning phase is the fundamental process of understanding why an information system should be built and determining how the project team will go about building it. It decide project goals and check feasibility (technical, economic, organizational).

Example in E-commerce platform: Decide to build an e-commerce website to sell electronics; check cost

and required technology.

② Analysis: The analysis phase answers the questions of who will use the system, what the system will do and where and when it will be used.

Example in E-commerce Platform: Identify that customers need product search, secure payment, order tracking; analyse competitor sites.

③ Design: The design phase decides how system will work $\rightarrow$ strategy (build, outsource, buy), architecture (hardware / software), database, programs.

Example in E-commerce Platform: Create database design for products & customers, plan website layout and payment gateway integration.

④ Implementation & Operation: The final phase of the SDLC build, testing, ~~impleme~~ installation, conversion, strategies (Parallel, Direct, phased, pilot).
Then ongoing operation + modifications.

Example in E-commerce platform: Developers code the catalog, shopping cart and payment modules; launch the site on a web server and later perform updates or bug fixes as users start using it.

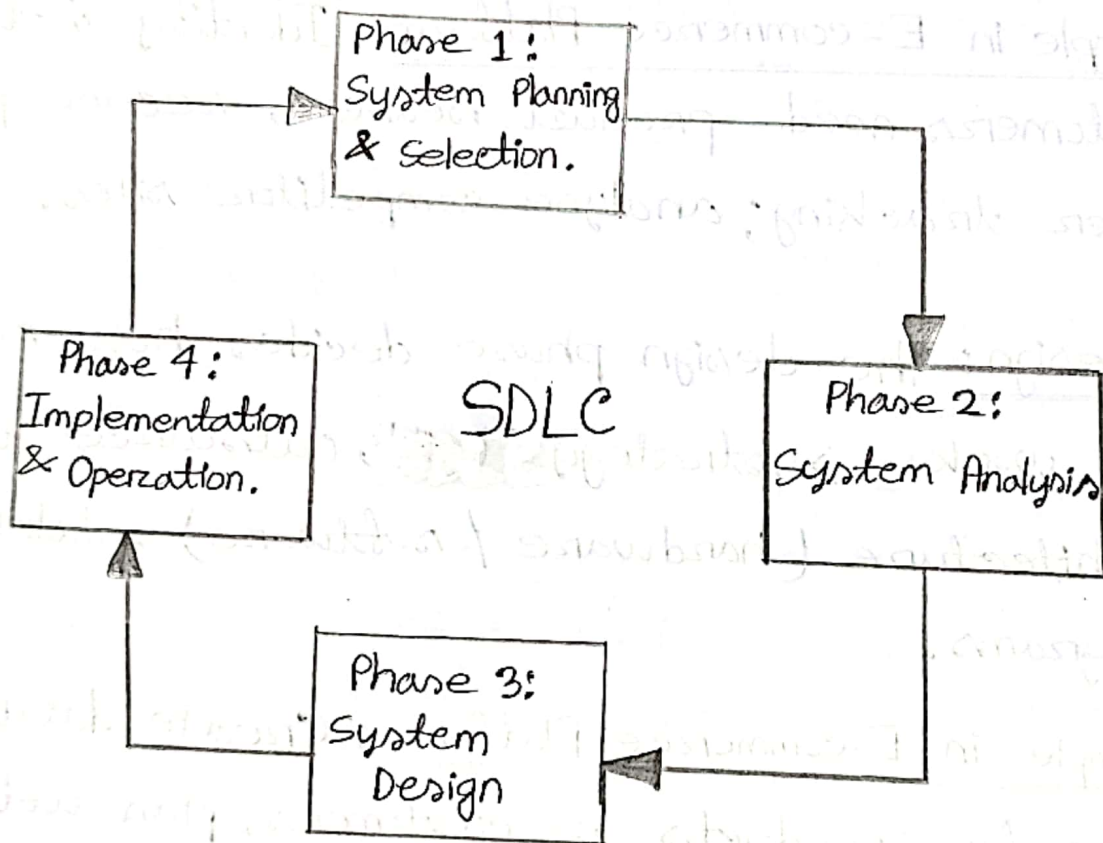


Figure: SDLC with four phases.

Ans. to the Ques. No — 2(b) → or

I would adopt the Waterfall SDLC model for developing an e-commerce platform that requires small and there are no uncertain requirements.

Reason:

- The e-commerce platform requirements are small and clearly defined.
- There is no uncertainty in requirements.
- Waterfall is best when the project is stable and well understood.

Advantages of waterfall for this case

- ① Simple and structured.
- ② Clear documentation at every stage.
- ③ Easy to manage because each phase must be completed before the next begins.
- ④ Works well when requirements will not change.

Disadvantages:

- ① Inflexible if requirements change later.
- ② Testing starts only after coding, so late detection of major errors is possible.

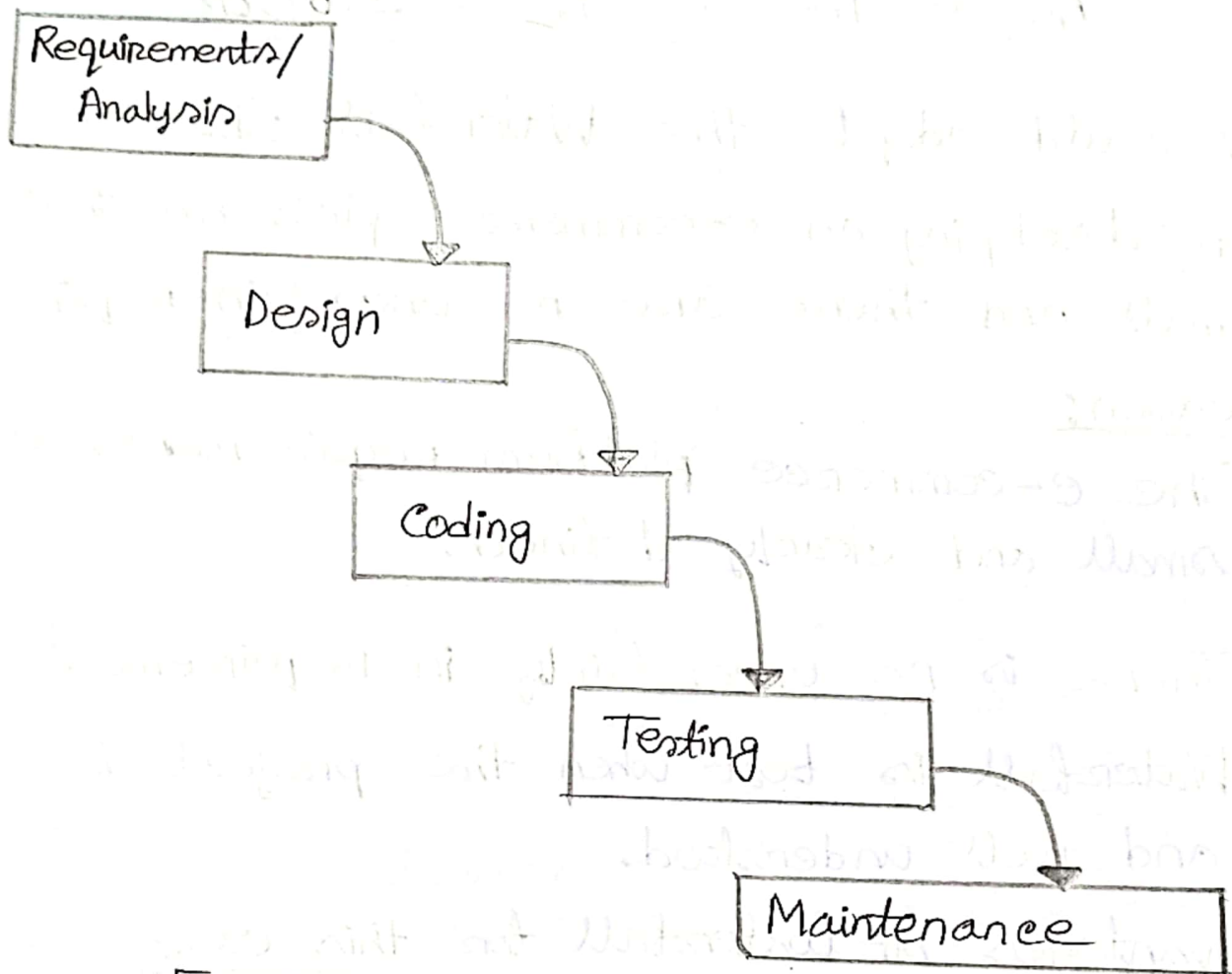


Figure: Waterfall Development Model.



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Ans. to the Ques. No-2(a)

Rapid Application Development (RAD):

RAD is a software development approach that focuses on fast development using prototyping, user feedback and CASE tools.

Advantages:

- ① Faster development process.
- ② Early user feedback ensures better alignment with needs.
- ③ Errors and issues can be identified early.

Disadvantages:

- ① Not suitable for large / complex projects.
- ② Less emphasis on long-term strategy.
- ③ Requires skilled team and proper tools.

Example: A Banking App is developed by first creating a basic prototype, then adding new features based on user feedback.

RAD Design Workshop

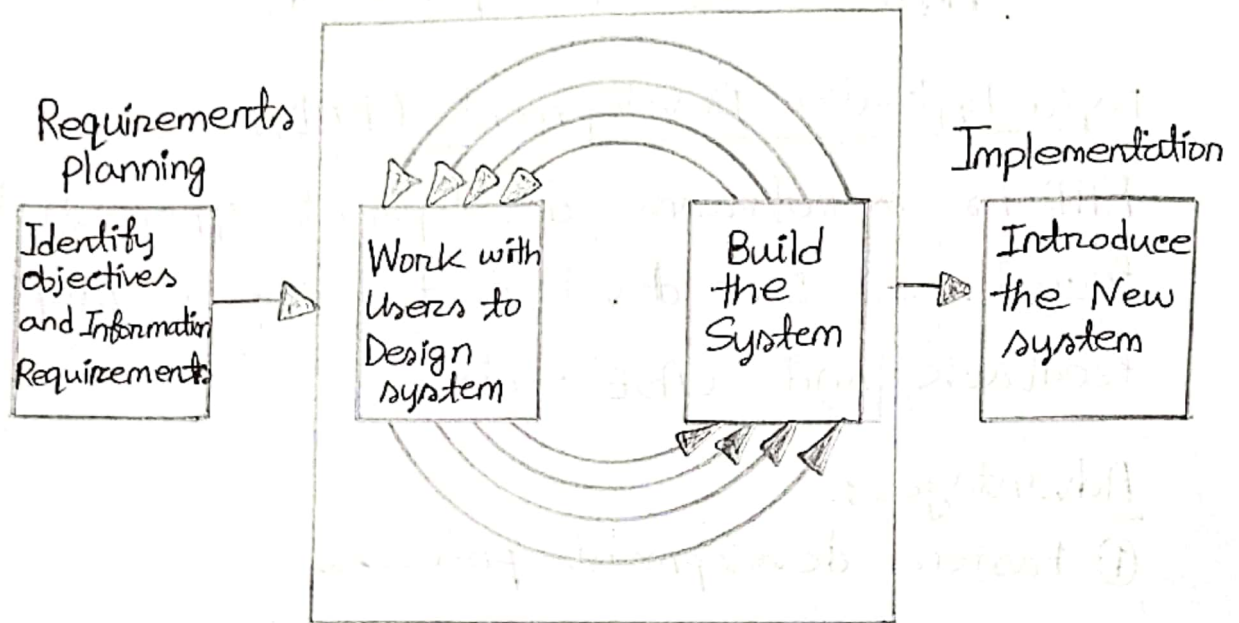


Fig: The four Phases of the RAD Model.

Joint Application Development (JAD):

JAD is a structured process where users, managers, and system analysts work together in workshops to define system requirements and design.

JAD = Workshop + Users + Managers + Analysts → Better Requirements.

Key Participants in JAD:

- Facilitator: Guides the entire session.
- Users: People who will use the system.
- Managers: Decision-makers focusing on business aspects.
- Analysts/Developers: Handle technical aspects.
- Scribes: Record notes and prepare documentation.

Advantage of JAD:

- ① Strong user involvement → system meets user needs.
- ② Faster requirement gathering.
- ③ Less miscommunication.
- ④ Conflicts resolved quickly through group discussions.

Disadvantage of JAD:

- ① Can be time-consuming (hard to gather everyone together).
- ② Requires a skilled facilitator.
- ③ Hard to apply in very large organizations.

Example: Imagine building a University Management System:

In a JAD workshop →

- Students (users): want online course registration.
- Teachers (users): want an easier grade submission system.
- University authority (managers): care about cost and security.
- Analysts: Record everything to form final requirements.

As a result, the system is developed based on everyone's input, making it more effective.

Ans. to the Ques. No - 2 (b)

Project ①: New Accounting Software

Recommended SDLC Model: Waterfall Model.

Justification:

The client has given detailed requirements at the beginning and does not expect major changes.

The project needs to be delivered quickly and in a fixed scope.

Waterfall is ideal because it is sequential

(Planning → Analysis → Design → Implementation/Operation) and works best when:

⇒ Requirements are well-defined and stable.

⇒ Clear documentation and structured phases save time.

Project ②: New Social Networking Platform

Recommended SDLC Model: Agile Model.

Justification:

A social networking site needs to adapt quickly to changing market trends and add features based on user feedback.

Use Agile for the social networking platform because it requires frequent updates, flexibility and user feedback to succeed in a fast-changing market.

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Repeat Question.

Ans. to the Ques. No-2(a) or

Developing an Online Learning platform recommended
SDLC Model : Agile

Justification:

The online learning platform needs continuous improvements - new subject modules, interactive tools, and usability updates. Educators give regular feedback, so requirements may change frequently as the platform grows.

Agile uses short iterations (sprints) to build and release features quickly, take user feedback and refine the system.

Advantages:

- ① Handles changing requirements easily.
- ② Delivers usable features early for testing by instructors and students.
- ③ Encourages strong collaboration between developers and educators.

Agile is best because it supports iterative development and regular feedback, ensuring the platform keeps improving based on educator and student needs.



Ans. to the Ques. No- 2 (b)

Distinguish between Methodologies, Models, Tools and Techniques ~~are~~ given below —

Methodologies	Models	Tools	Techniques
① It means overall framework or guideline that defines how to develop a system.	① It means graphical or mathematical representation of a system or process.	① It means software applications that support development work.	① It means specific method or skills applied in development.
② It focus process and steps to follow.	② It focus visual/abstract representation.	② It focus practical support for development tasks.	② It focus practical way of doing a specific task.
③ High-level approach	③ Conceptual/visual	③ Practical/Software-based	③ Skill/method-based
④ <u>Examples:</u> Waterfall, Agile, Spiral.	④ <u>Examples:</u> Flowchart, DFD, ERD, Gantt chart.	④ <u>Examples:</u> MS Project, Eclipse, MySQL, CASE tools.	④ <u>Examples:</u> Interviews, Normalization, Black-box testing, wireframing.